Hybrid Mechanical and Agentic Specification Inference

Wolfgang Grieskamp, Teng Zhang, and Vineeth Kashyap

Aptos Labs, Palo Alto, USA
wg@aptoslabs.com

Abstract. Formal verification suffers from the cost of specification: to establish high-level facts, contracts have to be written for supporting functions, many of them mechanical and low-level. Specification inference promises to automatically derive such contracts from the program. We present a hybrid of agentic and mechanical inference for Move, a smart contract language similar to Rust, integrated with the Move Prover. An agent controls inference and verification through a tool that, besides calling the Move Prover, derives specifications from code by weakest-precondition analysis (WP). We evaluate the approach on 26 non-trivial verification targets, 24 of them extracted from public Move code of the Aptos framework that has no specification, under a frontier model and a mid-tier model of the same family (Sol 5.6 and Terra 5.6). With and without WP, the agent completes over 90% of the tasks, so agentic inference is effective on its own; adding WP lowers its cost by 11–21% overall and by 40–64% on targets without loops.

Keywords: Specification inference · Weakest precondition · Reference elimination · LLM agents · Move Prover

1 Introduction

A roadblock to the adoption of program verification is the *cost of specification*. The properties developers care about are high-level — a call chain aborts only under stated preconditions, an entrypoint satisfies its postconditions, stored data respects an invariant — but to prove any of them, the supporting functions need contracts of their own, and those are often boilerplate, as they restate what the code does. *Specification inference* derives such contracts from the program, so that effort can go to the properties worth stating by hand. The classical engine for mechanical inference is Dijkstra's weakest-precondition transformation (WP) [12]: propagated backward through a function, it yields the exact abort condition and the exact postcondition. However, it depends on loop invariants to provide an induction hypothesis [8], and despite a long history of research [10], there is no complete technique for synthesizing them.

This paper describes a novel combination of agentic specification inference, as studied, e.g., in [26,16], and weakest-precondition transformation. The agent conducts the inference, supplies loop invariants, and calls WP and the verifier

as tools. The approach is implemented for the Move smart contract language, which is similar to Rust. Move has strong verification support through the Move Prover (MVP) [13,17]. Move shares its reference semantics with safe Rust and is therefore amenable to reference elimination, which the Move Prover has applied since its beginning [13]. On reference-eliminated Move bytecode, we run classical WP and map Move's references into the predicate calculus of the specification language. Moreover, *behavioral predicates* [17], which let a specification refer to the contract of another function as a black box, make WP fully modular.

We evaluate the combination on a corpus of 26 inference problems. Twenty-four of them are extracted from public Move code of the Aptos framework that has no specification: a model may have seen the code during training, but it cannot recall a specification that was never written. We compare three tactics, one without the WP tool and two with it, under two models of one family, the frontier model Sol 5.6 and the mid-tier model Terra 5.6, and check completeness with withheld mutants. Without the WP tool, the agent already does well: it completes 99% of its runs under the frontier model and 93% under the mid-tier model. With WP, the cost drops by 11–21% at about the same completeness. The saving comes from the targets without loops, where WP derives the complete contract and the hybrid tactics cost 36–60% of the agent alone; on loop targets, where the agent still has to find the invariant, they cost 9–22% more. The mid-tier model costs 53–59% of the frontier model, close to the ratio of their prices, and falls behind it mainly on the one target whose contract needs a permutation argument proved with lemmas; with WP, it completes 97 of 104 runs at 46% of what the frontier model spends without it.

Contributions. The paper makes the following contributions.

- A WP analysis over reference-eliminated bytecode, whose derived conditions are carried back to the references and global resources of the source, with state labels naming the intermediate memories of a function (Sec. 3).
- Diagnostics designed for an agent rather than a solver: loop-invariant evidence from bounded unrolling, and markers for conditions that are hard to satisfy or incomplete because of a callee (Sec. 3.5).
- The combination of this analysis with a coding agent, as a plugin of skills, hooks, and tools in which the agent supplies what WP cannot (Secs. 2 and 4).
- An evaluation on 26 targets from real-world code, whose completeness is scored with withheld mutants rather than by the agent or the authors, under three tactics and two models (Sec. 5).

The Move Prover was developed for the Diem project and has been maintained by Aptos Labs since 2022; it has verified Diem's payment system and critical components of the Aptos blockchain [24,2].

2 Hybrid Inference by Example

Hybrid inference combines two sources of specifications: the mechanical WP analysis of Sec. 3, which derives exact contracts but needs loop invariants to do

so, and a coding agent, which supplies the loop invariants and whatever else the analysis cannot derive, and repairs the specification based on verifier feedback. The prover is connected to the agent through a Model Context Protocol (MCP) service that offers a tool for verification and a tool for WP inference, and the agent carries a set of skills describing the language, the tools, and the workflow (Sec. 4). We walk through one session; the example is written in Move but should be readable without knowing the language [3].

The example computes the power $base^{exp}$ iteratively:

user

```
fun pow(base: u64, exp: u64): u64 {
  let result = 1;
  let i = 0;
  while (i < exp) {
    result = result * base;
    i = i + 1;
  }
  result
}
```

Following its skill instructions, the agent first calls WP. The answer reports the missing loop invariant and, as evidence, unrolls the loop symbolically and shows the values of the loop variables at each of the first loop entries:

wp tool

```
WP inferred ‘vacuous‘ conditions after this loop without an invariant. ..
  --> sources/pow.move:5:14
  |
5 |       while (i < exp) {
  |              ^
  |
  = loop-invariant evidence
  = bounded loop-head facts :
      head[0]: head[0].i = 0
               head[0].result = 1
      head[1]: 0 < exp ⟹ head[1].i = 1
               0 < exp ⟹ head[1].result = base
      head[2]: 1 < exp ⟹ head[2].i = 2
               1 < exp ⟹ head[2].result = base * base
      ..
```

From this evidence, the agent derives the loop invariant and introduces a recursive specification helper function for the accumulated result. It then calls the verification tool to check whether the invariant verifies:

agent

```
fun pow(base: u64, exp: u64): u64 {
  ..
  while (i < exp) {
```

```
      result = result * base;
      i = i + 1;
    } spec {
      invariant i ≤ exp;
      invariant result = math_pow(base, i);
    }
    ..
  }
  // specification helper function
  spec fun math_pow(base: num, exp: num): num {
    if (exp ≤ 0) 1
    else math_pow(base, exp - 1) * base
  }
```

In the next turn the agent runs the WP tool again. With the loop invariant present, WP derives the complete specification. However, it marks the abort condition as `sathard`. The marker is set when a derived condition is exact but its satisfiability is expected to be hard for the solver — here because of the existential quantifier, which states that the computation aborts if the multiplication overflows in some iteration `y`. The specification language of the Move Prover uses unbounded integers, `num`, which makes it possible to state overflow of the program's `u64` values:

wp tool

```
spec pow { // post and aborts condition for pow
  ensures result = math_pow(base, exp);
  aborts_if [inferred = sathard] ∃ y: u64:
    math_pow(base, y) * base > MAX_U64
    && (∀ x: num: 0 ≤ x && x ≤ y
          ⟹ math_pow(base, x) ≤ MAX_U64)
    && y < exp
}
```

The agent tries to verify this specification, but verification times out with the following diagnosis (the walkthrough runs with a solver timeout of 5 seconds, shorter than the 40 seconds of the evaluation):

verify tool

```
verification failed:
error: verification out of resources/timeout
       (timeout set to 5s)
 = timeout analysis replay:
     mixed quantifier and nonlinear arithmetic
 = top quantifier activity:
     quantifier at sources/pow.move:24
       -- 55,793+ observed instantiations
```

The quantifier the diagnosis points to is the recursive definition of the helper function `math_pow`, which the solver keeps unfolding while it tries to prove the `sathard` abort condition. In the next turn, the agent introduces lemmas that bound every partial product by the last one, which lets it restate the abort condition without a quantifier:

agent

```
spec pow {
  pragma opaque;
  ensures result = math_pow(base, exp);
  /// The partial products are non-decreasing, so the
  /// loop overflows exactly when its last product does.
  aborts_if
    exp > 0 && math_pow(base, exp - 1) * base > MAX_U64;
} proof {
  // Bounds every iteration's product by the last one;
  // triggers on the invariant's `math_pow(base, i)`.
  ∀ k: num {math_pow(base, k)}
    apply math_pow_last_is_max(base, k, exp);
}
spec lemma math_pow_last_is_max(
    base: num, k: num, n: num) {
  requires base ≥ 0 && 0 ≤ k && k < n;
  ensures math_pow(base, k) * base
            ≤ math_pow(base, n - 1) * base;
} proof {
  apply math_pow_mono(base, k + 1, n);
}
spec lemma math_pow_mono ...  // left out for brevity
```

This specification passes verification, and the agent returns it as the result after four turns.

3 Weakest Preconditions over Eliminated References

The mechanical half of the system is a weakest-precondition analysis [12] that reads off abort conditions, modifies clauses, and consequence-style postconditions from a function's bytecode. WP itself is textbook. Two things make it applicable to a language with Rust-style references and make its output readable: it runs on MVP's reference-eliminated bytecode, where backward predicate propagation is plain substitution, and a reverse mapping carries the derived predicates back to the references of the source.

3.1 Reference Elimination

Move has Rust-style references: `&T` for shared reads and `&mut T` for exclusive writes, both with a stack-bounded lifetime, and the borrow checker guarantees

that a live `&mut` is the only access path to its referent. MVP exploits this guarantee to rewrite bytecode into a form in which references are plain values [13], originally in order to keep the verification conditions first-order and free of a heap model. Two passes perform the elimination. Shared references are inlined: `&T` disappears and reads through it become reads of the underlying value. A mutable reference becomes a pair of the value it currently denotes and a *borrow path* back to its root, which is a local, a `&mut` parameter, or a global resource `R[a]` (Sec. 3.2); a write through the reference updates only this value. Where the lifetime of the reference ends, as determined by a borrow analysis, explicit *write-backs* are inserted along the path up to the first ancestor that is still alive; each rebuilds the parent value with the field or element replaced. A `&mut` parameter becomes an in/out value parameter: it enters as a plain value and leaves as the value written back last. Where the root of a reference is not statically unique, as in `if (c) &mut x else &mut y`, the write-back is guarded by an explicit test of which parent the reference was taken from. Before WP runs, the bytecode further passes through MVP's specification instrumentation: each function-level `requires`, `aborts_if`, `ensures`, or `modifies` clause and each loop `invariant` becomes an `assume` or `assert` at the appropriate program point, and calls are summarized by the *behavioral predicates* of [17]: `requires_of<f>`, `aborts_of<f>`, `ensures_of<f>`, `modifies_of<f>`, and `result_of<f>` reify a callee's contract as predicates over the call site's arguments and result. At the input to WP, mutable borrows, writes through references, and write-backs are still instructions of the bytecode, but they operate on values; all other instructions are assignments over plain values, interleaved with `assume` and `assert`.

3.2 Global Storage

In addition to Rust's reference semantics, Move has *global storage*, a memory indexed by a type and an address. A resource type `R` and an address `a` identify at most one value, written `R[a]`. The operations `move_to`, `move_from`, and `exists<R>(a)` publish, remove, and test this value, and `&R[a]` and `&mut R[a]` borrow it. The prover models global storage as one map from addresses to values per resource type, and the specification language uses the same notation `R[a]`, with `old(R[a])` for the value at function entry. Since the memory is indexed by type, accesses to different resource types never alias, and accesses to the same type alias only when their addresses coincide. A function that updates global storage several times passes through several memory states. The specification language can name such intermediate states with *state labels* and evaluate a condition at a labeled state `S`, written `S` $\mid\sim$ `e`; `old(R[a])` refers to the state at function entry.

3.3 The Predicate Transformer

The analysis is a backward dataflow pass over the control-flow graph. Its state at a program point is a triple (E, A, ℓ): a set E of *ensures* conjuncts that hold on every normal return reached from the point, a set A of *abort* disjuncts under which the function aborts, and a state label ℓ (Sec. 3.2) naming the memory state the

conditions refer to, together with bookkeeping for the `&mut` parameters and global resources that have been written on the way. At a return, $E = \{\, \texttt{result}_i = t_i \,\}$ for the returned temporaries and $A = \varnothing$; at an `abort`, $A = \{\texttt{true}\}$. Instructions transform the state as follows.

- *Assignment and operators.* $t := e$ substitutes e for t in E and A; an operator that can abort (arithmetic overflow and underflow, division by zero, an index out of range, a cast that truncates, a missing enum variant) also adds its abort predicate to A.
- *References.* The central instruction is the write-back, which ends the lifetime of a reference. A write-back of v into a parent `x` replaces `x` by `x` with the field or element of the edge set to v. A write $*r := v$ substitutes v for r, and a borrow $r :=$ `&mut x.f` substitutes `x.f` for r; together with the write-back, a borrow and the writes through it thus become a functional update of the root. A write-back to a global root `R[a]` instead adds `modifies R[a]` and an equation for the new value of `R[a]` at the current state label, and moves ℓ to the label before the write-back; the global borrow $r :=$ `&mut R[a]` substitutes `R[a]` for r and adds `!exists<R>(a)` to A. Since every reference is a path to a named root, no instruction needs a heap.
- *Calls.* A call $t := f(\bar{a})$ substitutes `result_of<f>`$(\bar{a})$ for t (or the callee's specification function, if f is pure), adds `aborts_of<f>`$(\bar{a})$ to A, and relates the memory before and after the call through `ensures_of<f>`; when the callee has no abort contract, the abort condition is recorded as *partial* rather than invented. Calls are thus summarized modularly: the caller's contract says what f returns without committing to how.
- *Branches.* At a branch on c the two incoming states are merged path-sensitively: conditions common to both sides are kept as they are, the others are guarded, $c \Rightarrow e$ and $\neg c \Rightarrow e'$ in E, $c \wedge a$ and $\neg c \wedge a'$ in A.
- *Specifications.* An `assert` φ adds φ to E and $\neg\varphi$ to A; an `assume` φ adds φ as an antecedent to both. This is how the contracts of callees, user-written conditions, and loop invariants enter the derivation.
- *Loops.* Loops are broken at their headers by the cut-point encoding of Barnett and Leino [8]: the back-edge is severed, the loop `invariant` is asserted before and after each iteration, and the loop-modified locations are havocked in between. WP of `havoc x` universally quantifies `x` in E and existentially in A, so variables that the invariant does not constrain appear quantified in the contract.

Since the remaining graph is acyclic, the pass visits each block once, in reverse topological order, and the state at entry is the raw contract. A post-processing step then maps it back to the source (Sec. 3.4) and simplifies it. Simplification matters more here than in a verifier, where the WP result is only passed to the solver: the raw predicate of a twenty-line function can have hundreds of terms, and the agent and the user read the simplified one.

3.4 Carrying the Result Back

The predicate at entry is stated over the reference-eliminated form, whose in/out parameters have no counterpart in the source signature; the contract the user reads must be stated over the original references. Three cases arise. A `&T` parameter needs nothing: the specification language already reads a shared reference as its value. For a `&mut T` parameter `p`, the specification language reads `p` as the value at exit and `old(p)` as the value at entry. The last value written to `p` becomes `ensures p = e`, where reads of `p` in `e` refer to `old(p)`. In straight-line code, several writes combine into one equation: `*p = *p + 1; *p = *p * 2` yields `ensures p = (old(p) + 1) * 2`. In a loop, `p` is havocked like any loop-modified variable, and the loop invariant has to relate it to `old(p)`. For a global resource, a borrow of `R[a]` that is written back yields `modifies R[a]` together with an equation on the resource's post-state value, with `old(R[a])` for its value at entry. Consider the transfer of an amount between two balances:

user

```
struct Balance has key { v: u64 }
fun transfer(from: address, to: address, amount: u64) {
  Balance[from].v -= amount;
  Balance[to].v += amount;
}
```

Both statements borrow a resource mutably, read and write a field through the reference, and write back. The derived contract is stated over the resources themselves, and the two borrow roots may coincide, so WP keeps them apart by a case split on `from = to`:

wp tool

```
spec transfer {
  pragma opaque;
  modifies Balance[from];
  modifies Balance[to];
  aborts_if !exists<Balance>(from);
  aborts_if !exists<Balance>(to);
  aborts_if Balance[from].v < amount;
  aborts_if
    from ≠ to && Balance[to].v + amount > MAX_U64;
  ensures
    from = to ⟹ Balance[from] = old(Balance[from]);
  ensures from ≠ to
    ⟹ Balance[from].v = old(Balance[from]).v - amount;
  ensures from ≠ to
    ⟹ Balance[to].v = old(Balance[to]).v + amount;
}
```

When a function updates global memory in several steps, an equation on the final memory is not enough: the abort condition of the second `move_to` below depends on the memory after the first. The state labels of the transformer therefore appear

in the contract, naming the intermediate memory states at which a global update happens; a condition is evaluated at a labeled state, `S |~ e`, or over a range of states, `S1..S2 |~ e`:

user

```
fun split_balance(
    dst1: &signer, dst2: &signer, src: address): u64 {
  let Balance { v } = move_from<Balance>(src);
  let half = v / 2;
  move_to(dst1, Balance { v: half });
  move_to(dst2, Balance { v: v - half });
  half
}
```

wp tool

```
spec split_balance {
  pragma opaque;
  modifies Balance[src];
  modifies Balance[signer::address_of(dst1)];
  modifies Balance[signer::address_of(dst2)];
  ensures result == old(Balance[src]).v / 2;
  ensures ..S1 |~ remove<Balance>(src);
  ensures S1..S2 |~
    publish<Balance>(signer::address_of(dst1),
      Balance { v: old(Balance[src]).v / 2 });
  ensures S2.. |~
    publish<Balance>(signer::address_of(dst2),
      Balance { v: old(Balance[src]).v
                   - old(Balance[src]).v / 2 });
  aborts_if !exists<Balance>(src);
  aborts_if
    S1 |~ exists<Balance>(signer::address_of(dst1));
  aborts_if
    S2 |~ exists<Balance>(signer::address_of(dst2));
}
```

`S1` names the memory between the `move_from` and the first `move_to`, `S2` the memory between the two `move_to`s; `publish<R>(a, x)` states that resource `R` at `a` equals `x` over the range, `remove<R>(a)` that none is present, and `exists<R>(a)` is the usual existence predicate. Without the labels, the memory versions the eliminated form threads through the function could be read back only as a single, weaker equation on the final memory.

3.5 Diagnostics for the Agent

WP is complete only where its inputs are: a loop without an invariant and a callee without an abort contract both leave holes, and the tool reports both instead of silently emitting an incomplete contract. After a havocked loop, a condition that

quantifies over everything the loop touched carries no information; such *vacuous* conditions are dropped and the loop is reported with *loop-invariant evidence* — the facts that hold at the first few loop heads on each path, computed by bounded unrolling of the same transformer, as in the `head[k]` listing of Sec. 2. A condition that is exact but existentially quantified, typically an abort condition that says *some* iteration overflows, is kept and marked `sathard`, since its satisfiability will be expensive for the solver. An abort condition that could not account for a callee is marked *partial*. Every emitted condition carries an `[inferred]` marker, so that re-runs replace their own output and never disturb user-written specifications (the listings in this paper omit the marker, except where it carries `sathard`). The agent's skills are written against these markers: they identify the loop that needs an invariant, the condition to simplify, and the callee to specify first.

3.6 What the Elimination Buys

Reference elimination was introduced to make *verification* of Move tractable [13]: without a heap, the verification conditions stay first-order and the solver needs no frame reasoning. A predicate transformer over programs with references has to carry a model of locations: `*r` means the contents of whatever `r` points to, an assignment through one reference may change the value read through another, and the transformer needs either an explicit heap with separation or ownership reasoning to keep the substitutions sound [23,18]. Its output is then stated over that model — heap fragments, prophecy variables, ownership predicates — and has to be translated before a user can read it. On the eliminated form none of this arises. Every reference is a path to a named root, so substitution is exact by construction; the only aliasing that survives is between global resources whose addresses may coincide, and it surfaces as an explicit condition such as `from == to` above. The derived predicates mention only parameters, results, fields, and resources, which is the vocabulary of the specification language, so the reverse mapping of Sec. 3.4 reduces to renaming. And because inference and verification run on the same pipeline, every inferred contract is checked under the semantics that produced it; the instrumented contracts of callees and the loop invariants are available in-line to both. The whole analysis is about nine thousand lines of Rust on top of the existing pipeline, by far the largest part of it being the simplification of predicates, and it adds no new semantics to the Move Prover.

4 The Agent Side

The agentic side of the workflow is a plugin for a coding agent: skills that instruct it, hooks that check generated Move as it is written, and the Move toolchain exposed via a local MCP server. The hooks run after every edit of a Move file: they parse the file, check its specification expressions, reject deprecated Move 1 constructs, and format the source, so that syntactic slips are caught before a verification run is spent on them. The plugin is rendered per tactic from one source; the setup is at [4], and the evaluation of Sec. 5 runs the rendered plugin

under the Codex CLI, which receives the skills inline with the task and the tools through the MCP server. We summarize the design.

Inference tactics. Inference runs in one of three *tactics*, which differ in whether the WP tool is available and who decides when to use it.

- *agent-only:* WP is unavailable. The contract and any loop invariants are derived directly from the implementation and the contracts of its dependencies, checked as the work proceeds.
- *hybrid-guided:* the workflow of Sec. 2, prescribed as a fixed order — run WP over the scope, take one warned function and write an invariant from the loop-invariant evidence (Sec. 3.5), rerun WP on that function alone until the warning is gone, then check.
- *hybrid-flexible:* WP is available as a tool, and the agent decides whether and when to call it and how to combine it with its own reasoning.

Skills. A tactic selects the *plan* — the sequence of tasks and which of them consult WP. All other guidance is shared reference material, identical in all three tactics: the specification language, the rules for writing and editing specs, the verification sub-workflow, and the toolchain's capabilities and limits. Only the description of the WP tool itself is withheld where the tool is absent; the underlying idea of backward reasoning, and what a loop without an invariant costs, are background that every tactic receives.

The most relevant areas of guidance are:

- *Loop invariants:* when to introduce recursive helper functions for iterative quantities, and when to prefer state-level invariants on resources — assumed by the prover at every call site, including each loop iteration — over per-loop inductive arguments.
- *Timeout resolution:* a hierarchy of escalations — strengthen state-level invariants, then introduce spec helpers, then lemmas, then explicit proof scripts — with a strict prohibition on weakening abort or postconditions to silence a timeout, since this gives up a real property only to make the proof succeed.

5 Evaluation

The evaluation asks what the mechanical component adds to an agent, in completeness and in cost, and whether it changes which model is needed. *Arm against arm* (Sec. 5.2) holds the model fixed and varies the tactic: *agent-only*, *hybrid-guided*, and *hybrid-flexible*, run under the same conditions. *Model against model* (Sec. 5.3) repeats the design under a frontier model and a mid-tier model of the same family and asks whether, with WP, the cheaper model reaches the results of the frontier model. Both rounds share the corpus, the protocol, the scoring, and the cost accounting described first.

5.1 Setup

Corpus. The corpus has 26 targets from 12 modules. Twenty-four are extracted from public Move code of the Aptos framework at a pinned commit: ten from its experimental trading modules, thirteen from modules for accounts, authentication, and rate and stake limits, and one from the Move standard library. The other two are written for the corpus. Function bodies are copied unchanged except for a few reductions documented in the module headers, and the package includes everything the targets depend on. No target has a specification upstream. Thirteen targets have a loop that needs an invariant. Each target has a hand-written reference specification that verifies within 20 seconds, and six *mutants* in two disjoint sets of three: patches to the implementation, one per obligation the contract has to pin, each shown to be killed by the reference. A task starts from the package with the reference contracts of every function the target calls; the contracts of the target and of all other functions are withheld, since a caller's contract can restate what the target does. With 3 arms and 4 replicates, a model round has **312 cells**.

Arms and models. The three tactics of Sec. 4 run as arms that differ only in the rendered plugin: the reference material is byte-identical, and in agent-only the WP tool is absent rather than discouraged. Two rounds use two models of one family through the same runtime, the Codex CLI at high effort: Sol 5.6, the frontier model, and Terra 5.6, the mid-tier model, priced at about half of Sol's list price. Each session has a budget of 100k output tokens, one hour of wall time, and 40 seconds of solver time per verification condition.

Scoring. A cell reaches *operational success* when a judge outside the agent's workspace confirms that the package compiles, the requested scope verifies, no weakening construct is present (`pragma verify = false`, partial abort contracts, vacuous postconditions, invented preconditions), the bytecode is unchanged, and the required contract categories are present. The agent's own claim is never a score, and the agent receives only this acceptance feedback: the mutants are withheld until generation has ended, and a contract is never handed back for strengthening. Operational success is necessary but weak: a vaguer contract passes it more easily. Completeness, postconditions as strong as the behavior, has no deductive check in general; WP guarantees it only given complete loop invariants and callee contracts. The second level is therefore *mutation adequacy*: after the round, each withheld mutant is applied to the implementation, and the prover is rerun against the inferred specification. A cell is *complete* when its specification kills all six mutants and *failed* when one survives; a cell in which a mutant reached no verdict within the solver's time limit is *unmeasured*. All 624 cells reached operational success, so the outcome of a cell is decided by the withheld mutants alone.

Cost accounting. Costs are in US dollars at each model's published list prices, computed from the recorded token counters of every model request; startup

Table 1. Both rounds, per arm: mean cost in US dollars per cell, the hybrid/agent-only cost ratio with its 95% bootstrap interval, complete, failed, and unmeasured cells, the share of cells whose first candidate check was accepted, mean input/output tokens in thousands, and mean wall time per cell in seconds.

model	arm	cost	vs. agent [95% CI]	compl.	failed	unm.	1st	I/O k	wall s
Sol 5.6	agent-only	$0.353	—	103/104	1	0	62%	333/3.8	121
	hybrid-guided	$0.304	0.86 [0.66, 1.09]	98/104	5	1	68%	288/3.5	126
	hybrid-flexible	$0.306	0.87 [0.74, 0.99]	98/104	2	4	71%	280/3.3	115
Terra 5.6	agent-only	$0.202	—	97/104	5	2	37%	367/3.9	153
	hybrid-guided	$0.161	0.79 [0.59, 1.02]	97/104	6	1	63%	289/3.1	148
	hybrid-flexible	$0.181	0.89 [0.73, 1.05]	98/104	6	0	65%	334/3.4	143

requests that do no task work are excluded. Per million tokens, Sol 5.6 is $4 uncached input, $0.40 cached input, and $20 output; Terra 5.6 is $2, $0.20, and $12. Thinking counts as output for both. Means are task-weighted, and the 95% intervals are percentile bootstraps over the 26 tasks (10,000 resamples). Differences in completeness are tested with the preregistered blocked randomization tests of the round analysis, Holm-adjusted.

5.2 Arm against Arm

Table 1 summarizes both rounds, Fig. 1 shows every cell, and Tables 2 and 3 in the appendix give every target.

Cost. Under both models the hybrids cost less than agent-only: 0.86× and 0.87× under Sol 5.6 and 0.79× and 0.89× under Terra 5.6 (hybrid-guided, hybrid-flexible). Over all targets, only the interval of Sol's hybrid-flexible excludes parity, but the targets split into two clear groups. On the 13 targets without a loop that needs an invariant, WP derives the complete contract and the agent only has to check it: the hybrids cost 0.36–0.60× agent-only, and every interval lies below 0.75. On the 13 loop targets they cost 1.09–1.22×: the agent still has to find the invariant, and the quantified WP output adds to the context it re-reads on every turn — on these targets the hybrids read 1.2–1.4× the input of agent-only (415–513k against 357–385k tokens per cell), on the loop-free ones 0.25–0.45×. Overall the hybrids read less input than agent-only (288k and 280k against 333k tokens per cell under Sol 5.6, 289k and 335k against 367k under Terra 5.6) and write less output.

Completeness. Measured completeness is about the same under all three tactics: agent-only completes 103 of 104 cells under Sol 5.6 and 97 under Terra 5.6, the hybrids 98 and 98, and 97 and 98, and under both models both preregistered contrasts have Holm $p = 1.00$. This is what one would expect: the loop invariants and specification helpers that decide the hard targets have to be found by the agent under every tactic. All failures but one Terra 5.6 agent-only cell occur

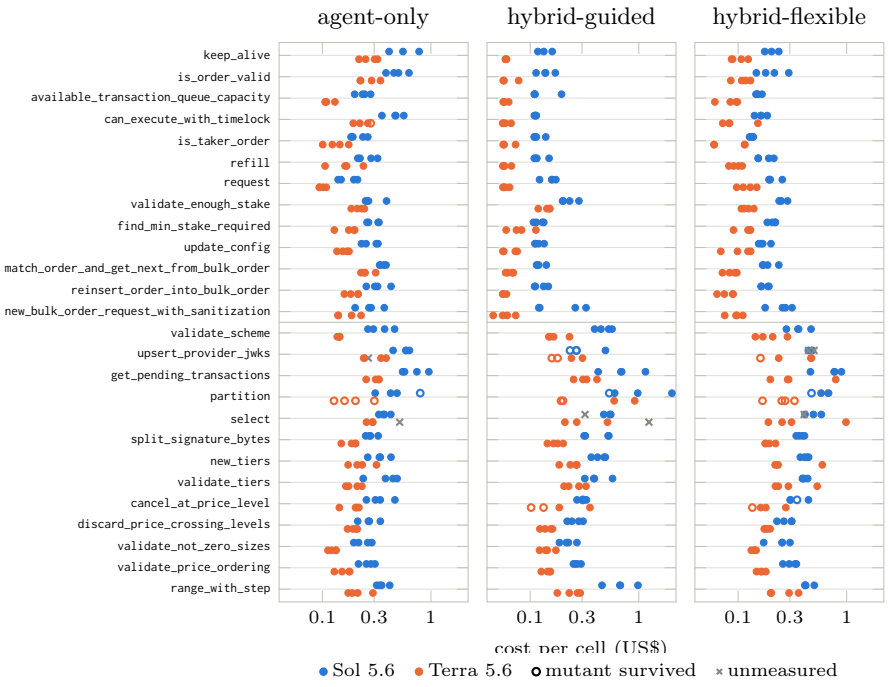


Fig. 1. Every cell of both rounds: cost in US dollars on a log scale, one panel per tactic, one row per target, and within each row the four Sol 5.6 replicates above the four Terra 5.6 replicates. The 13 targets above the line have no loop that needs an invariant. A filled dot is a complete cell, a hollow one a cell a withheld mutant survived, a cross an unmeasured cell.

on three loop targets. `partition`, an in-place partition, is the one target whose contract needs a permutation argument proved with lemmas; it fails in every arm, and in every failure a mutant that overwrites an element instead of swapping it survives: the contract does not state that the result is a rearrangement of the input. On the two scans `upsert_provider_jwks` and `cancel_at_price_level`, which leave the loop early, only hybrid cells fail (5 under Sol 5.6, 6 under Terra 5.6): the agent added invariants for the scanned prefix, accepted the quantified contract that WP derived from them once it verified, and a withheld mutant on the exit path survived. The hybrids do improve the first attempt: the share of cells whose first candidate check is accepted rises from 62% to 68–71% under Sol 5.6 and from 37% to 63–65% under Terra 5.6, and rejected checks drop from 66 to 40–60 and from 117 to 66–82.

5.3 Model against Model

Terra 5.6 costs 0.57× Sol 5.6 under agent-only and 0.53× and 0.59× under the hybrids, with all intervals within [0.45, 0.65], and it is cheaper on 25 or 26 of

the 26 targets in every arm. These ratios are close to the price ratio of 0.5 for input and 0.6 for output: the two models spend about the same number of tokens per cell, Terra 5.6 up to a fifth more input, and Terra 5.6 is slower (143–153 against 115–126 seconds). Completeness differs on one target. Outside `partition`, Terra 5.6 fails one agent-only cell and, under the hybrids, the same two scans as Sol 5.6; on the partition it fails 10 of 12 cells, Sol 5.6 3 of 12. The frontier model's advantage therefore lies in the lemma-based permutation argument; on the loop invariants, compositions, and global state that make up the rest of the corpus the two models are even. The one target on which Terra 5.6 costs about as much as Sol 5.6 is `select`, whose contract needs a recursive specification function over a function value; there it writes 1.5–2.0× Sol's output. With WP, Terra's first-check acceptance (63–65%) approaches Sol's (68–71%), and Terra 5.6 under hybrid-guided completes 97 of 104 cells at 46% of the cost of Sol 5.6 under agent-only, which completes 103.

6 Discussion and Conclusion

6.1 Interpreting the Results

Where WP helps. On targets without loops, WP needs no help: it derives the complete contract, and the agent only checks it, at 36–60% of the cost of writing the contract directly. Half of this corpus is of this kind, including four of the five compositions over callee contracts and eight of the nine targets with global state. On loop targets the agent still has to find the invariant, and WP's quantified output adds context that is re-read on every turn, so the hybrids cost slightly more there. With WP, the first candidate check is accepted more often, most for the mid-tier model (from 37% to 63–65% of cells).

Where it does not. The partition needs a permutation argument that WP cannot supply, and every tactic fails on it, the mid-tier model more often. The two scans that leave their loop early fail only in the hybrid tactics: the agent accepted the contract that WP derived from its prefix invariants once it verified, and a mutant on the exit path survived. A derived contract that verifies can thus still be incomplete, and the tool's diagnostics, which report a missing invariant, do not detect this case.

Mid-tier and frontier models. Both models spend about the same number of tokens per cell, so the cost of the mid-tier model follows its price. Apart from the permutation target, it handles the loop invariants, compositions, and global state of the corpus as well as the frontier model. We expect the cost advantage of WP to extend to cheaper models: for a local model, the context WP adds costs latency rather than money, and a model trained to read WP output should need fewer rounds with the tool.

Carrying over to prophecies. The reverse mapping of Sec. 3.4 relies on the elimination being *path-based*: an eliminated reference is still a path to a location the source can name, so the derived predicate has a source-level spelling. A *prophecy*-based elimination in the style of RustHorn and Creusot [23,11], in which a mutable reference is a pair of its current value and a prophecy variable for its final value, pinned when the borrow ends, has been designed for MVP but not implemented; it would need no borrow paths and would handle borrows through function values, which the path-based model cannot. A contract is model-agnostic, so inference could keep running on the path-based form regardless. Whether WP could run on the prophecy form itself and still be carried back is open. For a `&mut` parameter the prophecy *is* the post-state value and has the spelling `p`; for a reborrow through a function value the final value of the borrow has no source-level counterpart at all, and a predicate over it would have to be existentially closed or restated through the callee's contract. We do not know yet whether that always yields a readable contract.

Restrictions of validity. The reference specifications, the mutants, and the skills were written by the authors, who also built WP; the agent-only tactic receives the same reference material, and the mutants are the same for every tactic, but a different corpus or different skills could favor a tactic differently. Six mutants per target probe six obligations, so a contract that kills them all may still be weaker than the reference in ways the mutants do not test. Four replicates per cell bound what the comparison can resolve: the agent is not deterministic, and differences of a few cells per arm lie within that noise. Costs are list prices of one provider under one runtime, and the targets are functions of one codebase under two models of one family.

Open questions. Whether WP changes completeness is open. What we measured is about equal across tactics, and that is not surprising, since the loop invariants and specification helpers that decide the hard targets have to be found by the agent in every tactic; but the measure is coarse, and a difference small enough to hide behind six mutants and four replicates would still matter on a larger corpus. The scale of inference is open as well: the targets are functions, not modules. We plan to apply inference to the parts of the Aptos framework [2] that already carry hand-written specifications and to compare the inferred contracts with them — a comparison this corpus cannot make, since its targets were selected for having no specification.

6.2 Related Work

Specification inference predates large language models — Daikon [14] learned likely invariants from program traces, Houdini [15] kept the candidate annotations a verifier could not refute, and bi-abduction [9] inferred shape preconditions and postconditions compositionally — but these techniques stayed tied to particular property classes and source languages, and the advent of LLMs has revived interest

in the area. Pei et al. [25] showed that fine-tuned LLMs can predict program invariants statically at quality competitive with dynamic analysis. Kamath et al. [19] pair an LLM that *proposes* loop invariants with a symbolic checker that *validates* them, retrying on failure — a propose-and-validate loop that recurs in much of the subsequent work. For Verus, AlphaVerus [1] bootstraps verified code by self-improving translation with tree search over verifier feedback, and VeriStruct [26] synthesizes Verus-style specifications and proofs for Rust functions, modules, and data structures through planner-orchestrated phases. AutoSpec [27] generates ACSL specifications for C with an LLM guided by static analysis of the call graph and validated by Frama-C, and SpecGen [22] generates JML specifications for Java in a conversation with the verifier followed by mutation of the candidates. For smart contracts in Solidity, PropertyGPT [21] retrieves human-written Certora properties as input examples for specification generation. Most directly related to our work, MSG [16] targets the same Move specification language through a skill-based agentic design that consumes verifier feedback beyond a binary success/failure verdict. Across these systems the LLM remains responsible for the entire specification; the verifier serves as a binary oracle or, in MSG and our setup, as a feedback channel, but never as a contributor of mechanically derived content. A separate line of work, exemplified by DafnyPro [6] and AutoVerus [28], tackles the complementary problem of synthesizing the auxiliary proof annotations a verifier needs when the specification is already given. Our plugin can also synthesize proof hints (Sec. 2), but this is not the focus of this paper.

Reference elimination, the transformation underlying our WP analysis, was introduced in [13], and equivalents exist for Rust. Aeneas [18] compiles Rust to a pure functional model in a target prover (Lean, F*, Coq, HOL4) via forward and backward functions, while RustHorn [23] encodes a mutable borrow as a pair of its current value and a prophecy for its value at the end of the borrow, reducing verification to constrained Horn clauses. Neither infers specifications — in Aeneas the user writes them in the target prover, against the translated model — and both consume the reference-free form only in the forward direction. Our use of the transformation is the opposite: WP runs on the reference-free form, and the conditions it derives are carried *back* through the elimination. To our knowledge this inversion has not been described before. Boogie [7,20], the intermediate verification language that MVP translates to, also computes weakest preconditions of the procedures it is given [8] and passes them to the SMT solver; the difference is that Boogie's WP is not intended for external consumption, whereas ours has to be read, which is reflected in the reverse mapping and in the effort spent on simplification.

Two aspects thus distinguish our setup from the work above. First, to our knowledge, no prior LLM-based spec-inference system uses a sound mechanical baseline alongside the LLM. WP itself produces a substantial part of the inferred specification — abort conditions, modifies clauses, and consequence-style postconditions — so the AI is left only with what WP cannot recover, and the two streams are validated together by the Move Prover. Second, the workflow

is delivered as a plugin to a general-purpose, interactive coding agent via skills, edit-time hooks, and an MCP server, rather than as a custom-built tool: the agent participates in the loop and the user can intervene at any point.

6.3 Conclusion

We have presented a specification inference approach for the Move Prover that runs WP on reference-eliminated bytecode, where backward propagation is substitution and the derived contracts can be mapped back to the references of the source, and we have combined it with a coding agent that supplies what WP cannot: loop invariants, simplification, and proof hints. The prover checks both. Reference elimination was introduced for verification; this work shows that it serves inference as well, and the approach should carry over to verifiers for Rust-like languages that eliminate references along paths. On 26 targets, the agent alone already completes most cells, and WP lowers the cost wherever it derives the contract by itself, most for the cheaper model. We expect local and specialized models, for which the additional context costs least, to benefit further from WP.

Data Availability Statement. The corpus is published with the inference plugin at [4]: the package extracted from a pinned commit of the Aptos repository, the reference specifications, and both mutant sets. The round archives behind every table and figure — per-cell costs, token counters, wall times, and mutant outcomes — and the scripts that produce them from the archives are published there as well; [5] explains how to verify the corpus, re-score a round, and rerun the experiment.

Use of AI Tools. Parts of the text of this paper were drafted and edited with a coding agent (Claude Code); the authors reviewed and take responsibility for all content.

References

1. Aggarwal, P., Parno, B., Welleck, S.: AlphaVerus: Bootstrapping formally verified code generation through self-improving translation and treefinement. In: Proceedings of the 42nd International Conference on Machine Learning (ICML 2025). Proceedings of Machine Learning Research, vol. 267 (2025)
2. Aptos Labs: The Aptos Framework Book. https://aptos-labs.github.io/framework-book/ (2023), accessed: 2026-05-09
3. Aptos Labs: The Move on Aptos Book. https://aptos-labs.github.io/move-book/ (2023), accessed: 2026-05-09
4. Aptos Labs: MoveFlow: AI-assisted Move smart contract development. https://github.com/aptos-labs/aptos-core/tree/c1737fde43/aptos-move/flow (2026), commit `c1737fde43`, accessed 2026-10-06

5. Aptos Labs: Reproducing the corpus-v4 specification-inference results. https://github.com/aptos-labs/aptos-core/blob/c1737fde43/aptos-move/flow/evaluation/spec-inference/corpus-v4/REPRODUCE.md (2026), commit `c1737fde43`, accessed 2026-10-06
6. Banerjee, D., Bouissou, O., Zetzsche, S.: DafnyPro: LLM-assisted automated verification for Dafny programs (2026), https://arxiv.org/abs/2601.05385, dafny Workshop, POPL 2026
7. Barnett, M., Chang, B.Y.E., DeLine, R., Jacobs, B., Leino, K.R.M.: Boogie: A modular reusable verifier for object-oriented programs. In: International Symposium on Formal Methods for Components and Objects. pp. 364–387. Springer (2005)
8. Barnett, M., Leino, K.R.M.: Weakest-precondition of unstructured programs. In: Proceedings of the 6th ACM SIGPLAN-SIGSOFT Workshop on Program Analysis for Software Tools and Engineering. pp. 82—87. Association for Computing Machinery, New York, NY, USA (2005). https://doi.org/10.1145/1108792.1108813
9. Calcagno, C., Distefano, D., O'Hearn, P.W., Yang, H.: Compositional shape analysis by means of bi-abduction. In: Proceedings of the 36th ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages (POPL 2009). pp. 289–300. ACM (2009)
10. Cousot, P., Halbwachs, N.: Automatic discovery of linear restraints among variables of a program. In: Proceedings of the 5th ACM SIGACT-SIGPLAN Symposium on Principles of Programming Languages (POPL '78). pp. 84–96. ACM (1978). https://doi.org/10.1145/512760.512770
11. Denis, X., Jourdan, J.H., Marché, C.: Creusot: A foundry for the deductive verification of Rust programs. In: Formal Methods and Software Engineering (ICFEM 2022). LNCS, vol. 13478, pp. 90–105. Springer (2022). https://doi.org/10.1007/978-3-031-17244-1_6
12. Dijkstra, E.W.: Guarded commands, nondeterminacy and formal derivation of programs. Communications of the ACM **18**(8), 453–457 (1975). https://doi.org/10.1145/360933.360975
13. Dill, D.L., Grieskamp, W., Park, J., Qadeer, S., Xu, M., Zhong, J.E.: Fast and reliable formal verification of smart contracts with the move prover. In: Tools and Algorithms for the Construction and Analysis of Systems (TACAS 2022). Lecture Notes in Computer Science, vol. 13243, pp. 183–200. Springer (2022). https://doi.org/10.1007/978-3-030-99524-9_10
14. Ernst, M.D., Cockrell, J., Griswold, W.G., Notkin, D.: Dynamically discovering likely program invariants to support program evolution. IEEE Transactions on Software Engineering **27**(2), 99–123 (2001). https://doi.org/10.1109/32.908957
15. Flanagan, C., Leino, K.R.M.: Houdini, an annotation assistant for ESC/Java. In: Formal Methods Europe (FME 2001). Lecture Notes in Computer Science, vol. 2021, pp. 500–517. Springer (2001)
16. Fu, Y.F., Xu, M., Kim, T.: Agentic specification generator for Move programs. In: Proceedings of the 40th IEEE/ACM International Conference on Automated Software Engineering (ASE 2025). pp. 1286–1298 (2025). https://doi.org/10.1109/ASE63991.2025.00110
17. Grieskamp, W., Zhang, T., Kashyap, V., Silverman, J.: Formal verification of imperative first-class functions in Move. CoRR **abs/2605.10007** (2026), https://arxiv.org/abs/2605.10007, preprint; to appear in FMCAD 2026
18. Ho, S., Protzenko, J.: Aeneas: Rust verification by functional translation. Proc. ACM Program. Lang. **6**(ICFP), 116:1–116:31 (2022). https://doi.org/10.1145/3547647

19. Kamath, A., Senthilnathan, A., Chakraborty, S., Deligiannis, P., Lahiri, S.K., Lal, A., Rastogi, A., Roy, S., Sharma, R.: Finding inductive loop invariants using large language models (2023), https://arxiv.org/abs/2311.07948
20. Leino, K.R.M., Rümmer, P.: A polymorphic intermediate verification language: Design and logical encoding. In: Esparza, J., Majumdar, R. (eds.) TACAS. pp. 312–327. Springer Berlin Heidelberg, Berlin, Heidelberg (2010)
21. Liu, Y., Xue, Y., Wu, D., Sun, Y., Li, Y., Shi, M., Liu, Y.: PropertyGPT: LLM-driven formal verification of smart contracts through retrieval-augmented property generation. In: Network and Distributed System Security Symposium (NDSS 2025) (2025). https://doi.org/10.14722/ndss.2025.241357
22. Ma, L., Liu, S., Li, Y., Xie, X., Bu, L.: SpecGen: Automated generation of formal program specifications via large language models. In: Proceedings of the 47th IEEE/ACM International Conference on Software Engineering (ICSE 2025). IEEE (2025)
23. Matsushita, Y., Tsukada, T., Kobayashi, N.: RustHorn: CHC-based verification for Rust programs. ACM Trans. Program. Lang. Syst. **43**(4), 1–54 (2021). https://doi.org/10.1145/3462205
24. Park, J., Zhang, T., Grieskamp, W., Xu, M., Giacomo, G.D., Chen, K., Lu, Y., Chen, R.: Securing Aptos Framework with Formal Verification. In: 5th International Workshop on Formal Methods for Blockchains (FMBC 2024). Open Access Series in Informatics (OASIcs), vol. 118, pp. 9:1–9:16. Schloss Dagstuhl – Leibniz-Zentrum für Informatik (2024). https://doi.org/10.4230/OASIcs.FMBC.2024.9
25. Pei, K., Bieber, D., Shi, K., Sutton, C., Yin, P.: Can large language models reason about program invariants? In: Proceedings of the 40th International Conference on Machine Learning (ICML 2023). Proceedings of Machine Learning Research, vol. 202, pp. 27496–27520. PMLR (2023), https://proceedings.mlr.press/v202/pei23a.html
26. Sun, C., Sun, Y., Amrollahi, D., Zhang, E., Lahiri, S.K., Lu, S., Dill, D.L., Barrett, C.: VeriStruct: AI-assisted automated verification of data-structure modules in Verus. In: Tools and Algorithms for the Construction and Analysis of Systems (TACAS 2026). Lecture Notes in Computer Science, Springer (2026), https://arxiv.org/abs/2510.25015, arXiv:2510.25015
27. Wen, C., Cao, J., Su, J., Xu, Z., Qin, S., He, M., Li, H., Cheung, S., Tian, C.: Enchanting program specification synthesis by large language models using static analysis and program verification. In: Computer Aided Verification (CAV 2024). Lecture Notes in Computer Science, Springer (2024)
28. Yang, C., Li, X., Misu, M.R.H., Yao, J., Cui, W., Gong, Y., Hawblitzel, C., Lahiri, S.K., Lorch, J.R., Lu, S., Yang, F., Zhou, Z., Lu, S.: AutoVerus: Automated proof generation for Rust code. Proc. ACM Program. Lang. **9**(OOPSLA2) (2025). https://doi.org/10.1145/3763174

A Per-Target Results

Tables 2 and 3 give, for each target and arm, the mean cost per cell over the four replicates, mean input/output tokens in thousands (I/O k), the hybrid mean cost as a percentage of agent-only (vs. agent), and whether the specification killed every withheld mutant in all four replicates (✓) or a mutant survived (✗, subscripted with the count where more than one replicate failed); ∘ marks an unmeasured cell. Input counts include cached tokens. The targets above the middle rule have no loop that needs an invariant.

Table 2. Sol 5.6 by target. Summed over the 26 targets the arms cost $36.72, $31.67, and $31.81.

	agent-only			hybrid-guided				hybrid-flexible			
target	cost	I/O k	ok	cost	vs. agent	I/O k	ok	cost	vs. agent	I/O k	ok
keep_alive	$0.517	588/3.7	✓	$0.148	29%	69/0.7	✓	$0.206	40%	135/1.2	✓
is_order_valid	$0.493	577/3.7	✓	$0.140	28%	73/0.8	✓	$0.209	42%	166/1.3	✓
available_transaction_queue_capacity	$0.239	212/2.4	✓	$0.131	55%	66/0.7	✓	$0.153	64%	120/1.1	✓
can_execute_with_timelock	$0.465	454/4.9	✓	$0.112	24%	67/0.7	✓	$0.163	35%	114/1.0	✓
is_taker_order	$0.218	188/1.7	✓	$0.118	54%	66/0.6	✓	$0.133	61%	92/0.9	✓
refill	$0.260	217/3.6	✓	$0.121	47%	67/0.7	✓	$0.178	69%	135/1.1	✓
request	$0.174	129/1.7	✓	$0.153	88%	87/1.4	✓	$0.210	121%	160/2.0	✓
validate_enough_stake	$0.291	240/4.5	✓	$0.228	78%	154/2.5	✓	$0.255	88%	227/2.6	✓
find_min_stake_required	$0.296	282/2.3	✓	$0.121	41%	77/0.9	✓	$0.208	70%	157/1.8	✓
update_config	$0.280	285/2.8	✓	$0.119	42%	71/0.7	✓	$0.172	61%	149/1.1	✓
match_order_and_get_next_from_bulk_order	$0.358	332/4.2	✓	$0.123	34%	69/0.7	✓	$0.191	53%	131/1.0	✓
reinsert_order_into_bulk_order	$0.328	310/2.8	✓	$0.125	38%	67/0.7	✓	$0.177	54%	129/1.1	✓
new_bulk_order_request_with_sanitization	$0.280	213/3.3	✓	$0.207	74%	110/2.3	✓	$0.255	91%	160/2.9	✓
validate_scheme	$0.349	309/4.2	✓	$0.484	139%	486/6.7	✓	$0.366	105%	367/4.4	✓
upsert_provider_jwks	$0.565	574/7.5	✓	$0.314	56%	276/4.0	✗$_3$	$0.468	83%	422/7.9	○$_2$
get_pending_transactions	$0.705	779/6.6	✓	$0.739	105%	835/10.2	✓	$0.725	103%	833/8.7	✓
partition	$0.504	508/6.3	✗	$1.032	205%	1449/11.3	✗	$0.589	117%	658/7.2	✗
select	$0.374	273/5.6	✓	$0.472	126%	392/7.7	○	$0.475	127%	385/7.0	○
split_signature_bytes	$0.282	266/2.4	✓	$0.422	150%	465/4.3	✓	$0.378	134%	393/3.6	✓
new_tiers	$0.342	292/4.5	✓	$0.438	128%	388/6.5	✓	$0.414	121%	371/5.1	✓
validate_tiers	$0.389	418/3.8	✓	$0.416	107%	454/4.5	✓	$0.408	105%	472/4.1	✓
cancel_at_price_level	$0.342	266/4.0	✓	$0.303	88%	287/3.4	✗	$0.350	102%	264/3.6	✗
discard_price_crossing_levels	$0.272	235/2.7	✓	$0.262	96%	229/2.9	✓	$0.279	102%	261/3.1	✓
validate_not_zero_sizes	$0.240	197/1.5	✓	$0.224	93%	210/1.9	✓	$0.245	102%	261/2.1	✓
validate_price_ordering	$0.264	233/2.7	✓	$0.269	102%	253/3.1	✓	$0.309	117%	301/2.9	✓
range_with_step	$0.356	290/4.9	✓	$0.697	196%	711/10.2	✓	$0.438	123%	408/6.7	✓
all 26	$0.353	333/3.8	103/104	$0.304	86%	288/3.5	98/104	$0.306	87%	280/3.3	98/104

Table 3. Terra 5.6 by target. Summed over the 26 targets the arms cost $21.05, $16.71, and $18.79.

	agent-only			hybrid-guided				hybrid-flexible			
target	cost	I/O k	ok	cost	vs. agent	I/O k	ok	cost	vs. agent	I/O k	ok
keep_alive	$0.273	585/4.2	✓	$0.060	22%	69/0.4	✓	$0.102	37%	176/0.9	✓
is_order_valid	$0.269	574/3.7	✓	$0.062	23%	67/0.4	✓	$0.110	41%	154/0.9	✓
available_transaction_queue_capacity	$0.113	182/1.9	✓	$0.059	52%	71/0.4	✓	$0.085	75%	124/0.7	✓
can_execute_with_timelock	$0.239	470/4.4	✗	$0.059	25%	66/0.4	✓	$0.098	41%	165/0.9	✓
is_taker_order	$0.136	227/1.9	✓	$0.061	45%	66/0.4	✓	$0.088	64%	117/0.7	✓
refill	$0.168	267/3.9	✓	$0.060	35%	67/0.4	✓	$0.096	57%	141/0.9	✓
request	$0.100	151/1.6	✓	$0.059	59%	73/0.5	✓	$0.122	122%	171/1.9	✓
validate_enough_stake	$0.217	358/5.1	✓	$0.141	65%	239/2.4	✓	$0.121	56%	223/1.9	✓
find_min_stake_required	$0.175	321/2.3	✓	$0.083	47%	114/1.1	✓	$0.118	68%	201/1.7	✓
update_config	$0.159	289/2.7	✓	$0.065	41%	69/0.4	✓	$0.105	66%	175/0.9	✓
match_order_and_get_next_from_bulk_order	$0.257	517/4.8	✓	$0.065	25%	68/0.4	✓	$0.086	34%	127/0.8	✓
reinsert_order_into_bulk_order	$0.192	340/3.2	✓	$0.057	30%	66/0.4	✓	$0.079	41%	118/0.7	✓
new_bulk_order_request_with_sanitization	$0.173	263/3.8	✓	$0.059	34%	75/0.5	✓	$0.095	55%	141/0.8	✓
validate_scheme	$0.142	224/3.0	✓	$0.173	122%	292/3.5	✓	$0.202	142%	361/4.3	✓
upsert_provider_jwks	$0.310	543/8.1	○	$0.220	71%	346/5.1	✗$_2$	$0.335	108%	603/9.1	✗
get_pending_transactions	$0.306	663/5.1	✓	$0.325	106%	644/7.8	✓	$0.392	128%	850/9.2	✓
partition	$0.198	372/3.7	✗$_4$	$0.476	240%	1129/11.3	✗$_2$	$0.255	129%	489/5.8	✗$_4$
select	$0.338	612/9.0	○	$0.558	165%	1224/15.0	○	$0.435	129%	906/10.3	✓
split_signature_bytes	$0.185	351/2.8	✓	$0.172	93%	339/2.6	✓	$0.193	105%	355/3.7	✓
new_tiers	$0.232	412/5.9	✓	$0.238	102%	394/5.5	✓	$0.320	138%	660/7.4	✓
validate_tiers	$0.195	381/3.4	✓	$0.260	133%	595/4.8	✓	$0.320	164%	745/5.7	✓
cancel_at_price_level	$0.192	314/4.3	✓	$0.194	101%	344/4.3	✗$_2$	$0.188	98%	326/3.6	✗
discard_price_crossing_levels	$0.194	343/3.5	✓	$0.144	74%	220/2.5	✓	$0.184	95%	317/3.3	✓
validate_not_zero_sizes	$0.125	202/1.6	✓	$0.144	115%	225/1.7	✓	$0.140	112%	272/2.1	✓
validate_price_ordering	$0.159	255/2.2	✓	$0.144	91%	224/2.3	✓	$0.164	103%	293/2.8	✓
range_with_step	$0.216	331/5.4	✓	$0.242	112%	426/5.8	✓	$0.264	122%	489/6.7	✓
all 26	$0.202	367/3.9	97/104	$0.161	79%	289/3.1	97/104	$0.181	89%	335/3.4	98/104